

Patient Profile

Name: Yash M. Patel

Age: 21 years

Gender: Male

Summary of Results

The predictive model evaluates this laboratory profile as **within normal limits**.

Notable individual findings: elevated alanine aminotransferase (ALT) and reduced albumin and alkaline phosphatase (ALP).

Detailed Test Explanations

- **AST (SGOT)** – 16 U/L (15–40 U/L): within normal limits; indicates no significant hepatic or muscular injury.
- **ALT (SGPT)** – 100.50 U/L (10–49 U/L): markedly elevated; suggests hepatocellular damage or inflammation.
- **AST:ALT Ratio** – 0.50 (<1.0): normal pattern; with isolated ALT elevation, this may reflect acute or early chronic liver injury rather than alcoholic liver disease.
- **GGTP** – 10.20 U/L (0–73 U/L): normal; indicates no cholestasis or biliary obstruction.
- **Alkaline Phosphatase (ALP)** – 15.40 U/L (30–120 U/L): low; may reflect decreased bone turnover or nutritional deficiency.
- **Bilirubin Total** – 0.60 mg/dL (0.30–1.20 mg/dL): normal; no overt jaundice.
- **Bilirubin Direct** – 0.10 mg/dL (<0.3 mg/dL): normal; no conjugated hyperbilirubinemia.
- **Bilirubin Indirect** – 0.10 mg/dL (<1.10 mg/dL): normal.
- **Total Protein** – 6.39 g/dL (5.70–8.20 g/dL): within reference range.
- **Albumin** – 2.00 g/dL (3.20–4.80 g/dL): markedly low; indicates impaired synthetic function or protein loss.
- **A:G Ratio (Albumin:Globulin)** – 0.10 (0.90–2.00): extremely low, reflecting disproportionately low albumin relative to globulin.

Overall Interpretation

The laboratory data reveal a pronounced elevation of ALT, a low serum albumin concentration, and a reduced A:G ratio, all of which may point toward early or subclinical hepatic dysfunction. The low ALP could signify decreased bone activity or a nutritional issue. Despite these isolated abnormalities, the comprehensive predictive model still classifies the overall health profile as **normal**. Clinical correlation is recommended to determine the significance of the individual abnormal results.

Recommendations

1. Liver Evaluation

- Repeat liver enzymes (ALT, AST, GGT) in 4–6 weeks to assess trend.
- Consider hepatitis serology and imaging if ALT remains elevated.

2. Nutrition and Protein Status

- Evaluate dietary intake for adequate protein and essential amino acids.
- Assess for possible protein loss (e.g., gastrointestinal or renal) or malabsorption.

3. Bone Health

- Review calcium, vitamin D, and phosphate levels; assess for dietary deficiencies.

4. Lifestyle Modifications

- Limit alcohol and avoid hepatotoxic medications.
- Maintain a balanced diet rich in fruits, vegetables, and lean proteins.

5. Follow-up

- Re-examine liver panel and albumin in 3 months.
- If abnormalities persist or worsen, refer to a hepatologist.

Disclaimer

This report is for informational purposes only and does not replace individualized medical advice from a qualified healthcare professional.